

tree diagram to compute your expected return. The tree diagram is shown in Figure 18.6.

The "payoff" values in Figure 18.6 are computed by dividing the \$6 pot¹ among those players who guessed correctly and then subtracting the \$2 that you put into the pot at the beginning. For example, if all three players guessed correctly, then your payoff is \$0, since you just get back your \$2 wager. If you and Nick guess correctly and Eric guessed wrong, then your payoff is

$$\frac{6}{2} - 2 = 1.$$

In the case that everyone is wrong, you all agree to split the pot and so, again, your payoff is zero.

To compute your expected return, you use equation (18.3):

$$\begin{aligned} \text{Ex}[\text{payoff}] &= 0 \cdot \frac{1}{8} + 1 \cdot \frac{1}{8} + 1 \cdot \frac{1}{8} + 4 \cdot \frac{1}{8} \\ &\quad + (-2) \cdot \frac{1}{8} + (-2) \cdot \frac{1}{8} + (-2) \cdot \frac{1}{8} + 0 \cdot \frac{1}{8} \\ &= 0. \end{aligned}$$

This confirms that the game is fair. So, for old time's sake, you break your solemn vow to never ever engage in strange gambling games.

The Impact of Collusion

Needless to say, things are not turning out well for you. The more times you play the game, the more money you seem to be losing. After 1000 wagers, you have lost over \$500. As Nick and Eric are consoling you on your "bad luck," you do a back-of-the-envelope calculation and decide that the probability of losing \$500 in 1000 fair \$2 wagers is very, very small.

Now it is possible of course that you are very, very unlucky. But it is more likely that something fishy is going on. Somehow the tree diagram in Figure 18.6 is not a good model of the game.

The "something" that's fishy is the opportunity that Nick and Eric have to collude against you. The fact that the coin flip is fair certainly means that each of Nick and Eric can only guess the outcome of the coin toss with probability 1/2. But when you look back at the previous 1000 bets, you notice that Eric and Nick never made the same guess. In other words, Nick always guessed "tails" when Eric guessed "heads," and vice-versa. Modelling this fact now results in a slightly different tree diagram, as shown in Figure 18.7.

¹The money invested in a wager is commonly referred to as the *pot*.